Humans Disengage, Reasoning Models Persist: An Item-Controlled Dissociation in Deliberation Allocation

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Abstract

Thinking large reasoning models (LRMs) take longer on harder problems, just as humans do. This surface similarity hides an opposite pattern within items. When an LRM gets a problem wrong, it spends more tokens on it than when it gets the same problem right. Humans do the reverse: they spend less time on the trials they get wrong. We document this in a public matched human–LRM corpus, treating cross-item alignment with difficulty (registration) and the within-item wrong-vs-right contrast (allocation) as two levels of deliberation. On H-ARC, every analysed LRM has a wrong-vs-right effect of Cohen's $\boldsymbol{d = 1.47}$–$\mathbf{3.13}$; humans, holding item identity fixed, have the opposite sign. The dissociation survives item fixed effects, replicates on INTUIT, and weakens but does not invert on Cortes. We read the human pattern as engagement vs. abandonment: people stay on items they expect to solve and give up on items they don't. We read the LRM pattern as length on uncertainty: longer chains when the model is unsure, which is exactly when it tends to fail. Both policies produce the same cross-item correlation with difficulty, which is what previous work has measured; they come apart only within items. The result motivates a two-level computational-behavioral diagnostic that separates cross-item difficulty registration from within-item deliberation allocation, reframing human–LRM alignment as a question about stopping and control policies rather than trace length alone.

Keywords: large reasoning models, deliberation, reaction time, chain-of-thought, resource-rational analysis

Statements and Declarations

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Author Contributions. H. Wang is the sole author of this manuscript and is solely responsible for the conception of the study, the design and implementation of the analyses, the interpretation of the results, and the writing of the manuscript.

Ethics Approval. This study did not involve any new collection of data from human or animal participants. All analyses are secondary re-analyses of publicly released, fully de-identified behavioural and large reasoning model output corpora (the de Varda et al. release, H-ARC, INTUIT, and Cortes); no identifiable personal data are accessed at any point in the pipeline. Under the institutional research-ethics policy of The University of Hong Kong and the Common Rule (45 CFR 46.104(d)(4)), secondary analysis of publicly available de-identified data is exempt from ethics-board review. No further ethics approval was therefore required or sought.

Consent to Participate. Not applicable. No new human participants were recruited or tested for this study; the analyses use only publicly released, de-identified secondary data, for which the original data collectors obtained the relevant participant consent.

Consent to Publish. Not applicable. The manuscript contains no individually identifiable data, images, or other personal information from any human participant.

Data and Code Availability. All behavioural data analysed in this study are public and are not redistributed by the present manuscript. The de Varda et al. release is at https://osf.io/r3kum/; H-ARC is accompanied by LeGris, Vong, Lake, and Gureckis (2025); INTUIT is accompanied by Prunty, O'Flynn, Quinn, and Cheke (2025); Cortes is redistributed in the de Varda deposit. The full analysis pipeline (data acquisition, scoring, statistical analyses, and figure generation), the intermediate result CSVs, and the rendered manuscript figures are openly available at the OSF repository osf.io/jxb4a (anonymised view-only link for review), deposited concurrently with submission. The repository ships pinned dependencies, a deterministic random seed (`0xA12C`), and a `Makefile` target that reproduces every results CSV and figure end-to-end from the public source data. The repository will be de-anonymised and assigned a permanent DOI (Zenodo) at acceptance.

1 Introduction

A central question in the cognitive science of deliberation is how thinkers decide when an item is worth more thought, and when it is not. Resource-rational analysis casts the decision as a comparison of expected gain to opportunity cost (Callaway et al., 2022; Gershman, Horvitz, & Tenenbaum, 2015; Lieder & Griffiths, 2020; Russell & Wefald, 1991; Simon, 1956). Metacognitive theories locate it in confidence-based monitoring and disengagement (Fleming & Lau, 2014; Nelson & Leonesio, 1988; Yeung & Summerfield, 2012). Sequential-sampling models locate it at the boundary that terminates evidence accumulation (Bogacz, Brown, Moehlis, Holmes, & Cohen, 2006; Heitz, 2014; Ratcliff & McKoon, 2008). All three traditions converge on a two-step picture. An agent first *registers* that an item is hard — a perceptual or evaluative difficulty signal. The agent then *allocates* computation around that registered difficulty: a stopping or scheduling rule that decides whether to keep thinking on the item in front of it. Difficulty registration and deliberation allocation are dissociable: an agent can grade items by difficulty in the same order as another agent yet schedule its own computation on a different policy.

Large reasoning models (LRMs) make this decomposition empirically tractable. They extend the chain-of-thought paradigm of producing intermediate reasoning before a final answer (Kojima, Gu, Reid, Matsuo, & Iwasawa, 2022; Snell, Lee, Xu, & Kumar, 2025; Wei et al., 2022), and the cognitive-psychology-style analysis of language models has begun mapping their behavioural signatures against human ones (Binz & Schulz, 2023). Like a human reaction time, an LRM's reasoning-trace length is a behavioural readout of how long the agent dwells on a problem; the two measures are analogous, not interchangeable. de Varda, D'Elia, Kean, Lampinen, and Fedorenko (2025) reported a striking alignment on this measure: across seven reasoning tasks, LRM reasoning-token length tracks human reaction time both within and across paradigms. Items that take humans longer also elicit longer LRM traces. That result is a clean demonstration of difficulty registration. It is silent on allocation: cross-item correlations describe how the two systems grade items relative to one another, not how either system schedules computation around its own successes and failures.

The published commentaries on de Varda et al. (2025) press exactly this gap. Vankov, Adolfi, Heaton, Puebla, and Bowers (2026) questions whether trace length is a meaningful analogue of reaction time at all. Dujmović (2026) argues that the cross-item correlation is a difficulty-confound rather than a signature of shared mechanism, and Hu (2026) raises related concerns about what is being aligned. The authors' replies (de Varda, D'Elia, Kean, Lampinen, & Fedorenko, 2026a, 2026b) are themselves revealing: they reframe the original finding as "an empirical phenomenon worth explaining" rather than a mechanistic claim of shared reasoning, and explicitly defer the allocation question. The literature has thus converged on three points: cross-item alignment is real, it is not by itself a mechanism claim, and a separate diagnostic is needed to test how deliberation is allocated.

We supply that diagnostic. The outcome-conditioned within-agent question is simple: does the agent spend more deliberation on trials it answers correctly than on trials it answers incorrectly? We ask it for each agent on each paradigm, and we then use item fixed effects to test the human–LRM interaction. We use "within-agent" in the behavioral-comparison sense that deliberation is normalised and outcome-conditioned within each agent's own scale; for the item-fixed LRM slope, the public data identify an item-controlled contrast across models attempting the same item, rather than across repeated stochastic samples from a single model. The diagnostic is therefore better described as item-controlled than as truly within-agent in the repeated-measurements sense; we retain the term for continuity but flag the identification scope explicitly here and in Limitations. Each agent serves as its own baseline, so seconds and tokens are never compared directly. The contrast is a slope on a within-agent observable, and item fixed effects absorb the cross-item difficulty gradient that the original de Varda et al. (2025) finding lives on. The within-agent slope is therefore a diagnostic of deliberation *allocation* that is orthogonal to, and complements, the cross-item difficulty-registration diagnostic (Figure 1). The two diagnostics together form a minimal two-level test of human–LRM alignment in deliberation.

The scope is bounded. The LRM sample is six open-weight thinking models with DeepSeek-V3 as a non-thinking baseline; closed-frontier reasoning systems are not tested. The human comparison is to the samples reported in the source release rather than humans in general. Three reasoning paradigms are analysed: H-ARC (visual abstraction) as the primary paradigm, INTUIT as a structurally distinct cross-paradigm replication, and Cortes (binary relational reasoning) as a paradigm-dependent boundary case. The Methods, Results, and Discussion that follow develop the diagnostic, report its outcome, and recover from the outcome a candidate reading of what the two systems are doing differently when they think.

Contributions.

This paper makes three contributions. First, it separates two behavioral levels that are routinely conflated in human–model comparison: cross-item difficulty *registration* (does deliberation length track which items are hard?) and within-item deliberation *allocation* (given items recognised as hard, where does the agent spend more thought?). Second, it introduces an outcome-conditioned, item-controlled diagnostic that asks whether an agent allocates more deliberation to its own successes or failures, using each agent's native deliberation scale and holding item identity exactly fixed. Third, it uses this diagnostic to show that apparent human–LRM alignment at the cross-item level masks an opposite within-item allocation policy. The point is not that seconds and reasoning tokens are equivalent, nor that reasoning traces directly reveal latent computation. Rather, the point is that both are observable behavioural readouts whose outcome-conditioned structure, when read within each agent's own scale, constrains theories of metacognitive control, stopping, and resource allocation in a way that cross-item correlation alone cannot.

2 Methods

Datasets and models.

This study reanalyses publicly released, de-identified human and LRM data; no new human-subjects data were collected and no IRB review was required. We use public data from the de Varda et al. (2025) release and the underlying behavioural corpora (Cortés et al., 2021; LeGris et al., 2025; Prunty et al., 2025). Table 1 summarises the three reasoning paradigms. H-ARC (Chollet, 2019; LeGris et al., 2025) is the primary paradigm. INTUIT is the clean cross-paradigm replication; Cortes is a boundary case (*Generalisation and the Cortes boundary*); arithmetic is excluded because all thinking LRMs are at ceiling. The thinking-LRM sample (DeepSeek-R1 (Guo et al., 2025), Qwen-QwQ-32B and Qwen3-235B-Thinking (Qwen Team, 2025), GLM-4.5-Air-FP8 (Z.ai, 2025), gpt-oss-20b and gpt-oss-120b (OpenAI, 2025)) is open-weight only. DeepSeek-V3 (DeepSeek-AI, 2024) is a non-thinking control. Within H-ARC four LRMs are well-powered (DeepSeek-R1, GLM-4.5-Air-FP8, gpt-oss-120b, Qwen-QwQ-32B; all $n_{\mathrm{LRM}} \geq 298$); gpt-oss-20b ($n = 119$, parser-limited) and Qwen3-235B-Thinking ($n = 43$, 4 wrong trials) are reported descriptively. All six thinking LRMs enter on INTUIT and Cortes.

Deliberation measures.

For thinking LRMs, deliberation t is the `reasoning_token_length` field of the released frame: the number of intermediate-trace tokens generated before the final answer, counted in each model's own native tokenizer. For the non-thinking DeepSeek-V3 control we use `total_output_tokens`. We treat t as a behavioural readout of the model's generation policy under nominally-greedy decoding, not as a meter of latent computation: trace tokens are not guaranteed reports of underlying reasoning (Kambhampati et al., 2025; Samineni, Kalwar, Gangal, Bhambri, & Kambhampati, 2025; Valmeekam, Stechly, Palod, Gundawar, & Kambhampati, 2025). Each agent is its own within-agent baseline, so seconds and tokens are never compared directly. Following de Varda et al. (2025)'s analysis script, we deduplicate the released frame to one (item, model) row keeping the first attempt. INTUIT traces are restricted in the public release, so *Inside the dissociation* is restricted to H-ARC and Cortes. Full provenance, dedup, and per-model parseable/correct/wrong counts are in the SI (*S6*).

Paradigm	Cognitive domain	Human data	LRM data
H-ARC	Visual abstraction	400 items; 4,091 trials	5 thinking LRMs in within-agent d-ratio summary; 6 in difficulty-controlled regression[a]
INTUIT	Intuitive physics	144 items; 1,536 trials	6 thinking LRMs with both outcomes
Cortes	Relational reasoning	86 items; 20,052 trials	6 thinking LRMs (5–8 wrong trials each)
Arithmetic	Parametric arithmetic	168 items; 1,680 trials	all LRMs at ceiling (excluded)

Table 1: Datasets analysed. [a] Qwen3-235B-Thinking ($n = 43$ matched H-ARC items) and gpt-oss-20b ($n = 119$, parser-limited) are flagged as low-power on H-ARC and reported separately. Full dataset and scoring detail are in the OSF deposit.

The within-agent d-ratio.

For agent A, paradigm P, trial i with deliberation $t_i > 0$ and correctness $c_i \in \{0, 1\}$, the d-ratio is the pooled-SD Cohen's d on log deliberation. Here t_i is reaction-time seconds for humans, `reasoning_token_length` for thinking LRMs, and total output length for V3: $d_{\text{wrong}-\text{right}}(A, P) = (\bar{\ell}_{\text{wrong}} - \bar{\ell}_{\text{right}})/s_{\text{pooled}}$ with $\ell_i = \log t_i$. Positive values mean the agent deliberates longer on its failures. The contrast is symmetric: each agent is its own baseline, so seconds and tokens are never compared directly.

Difficulty controls.

A positive d-ratio could in principle reflect a value-insensitive stopping rule or a difficulty effect that correlates with errors. To break the tautology we use two within-agent controls. (a) A leave-one-out ensemble-difficulty regression $\log t_i = \alpha + \beta_{\text{correct}}\, c_i + \beta_{\text{diff}}\, D_i^{(-A)} + \varepsilon_i$, where $D_i^{(-A)} = 1 - \bar{c}_{i,-A}$ and $\bar{c}_{i,-A}$ is the mean correctness on item i taken over all non-focal agents available for that paradigm. When the focal agent A is the human population, the non-focal agents are all thinking LRMs that attempted item i. When A is one of the thinking LRMs, the non-focal agents are the human population plus every other thinking LRM that attempted item i. (b) Item fixed-effects regressions that absorb item identity with item dummies, identifying the correctness coefficient purely from variation within an item. The headline combined dissociation specification is $\log t_i = \alpha + \beta\, c_i + \beta_{\text{LRM}}\, c_i \times \text{LRM}_i + \delta_{\text{agent}} + \gamma_{\text{item}} + \varepsilon_i$ pooling humans and LRMs; the interaction β_{LRM} is the within-item dissociation. We additionally fit humans-only and LRMs-only item-FE specifications, per-LRM versions of the combined regression (humans plus one LRM at a time), the same three regressions on INTUIT and Cortes, and a Mundlak (Mundlak, 1978) re-parameterisation that replaces item dummies with item-level mean correctness. Cluster-robust SE by item throughout. Full specifications and analysis scripts are in the OSF deposit. **Identification scope.** The released LRM data contain exactly one observed outcome per (item × model) pair, so the within-item LRM correctness slope in the LRMs-pooled and per-LRM combined regressions is identified by variation across LRMs (and across the human population) on the same item, not by repeated stochastic samples of a single model. The interaction β_{LRM} is therefore an item-controlled human–LRM contrast, not a stochastic per-model slope; a stochastic single-model estimate would require multiple LRM samples per item, which the public release does not provide. Full specification rationale is in SI *S7*.

Hierarchical mixed-effects test on H-ARC.

We fit $\log t \sim c_c + \text{LRM} + c_c \times \text{LRM}$ with centred correctness, LRM indicator, and an item random intercept. A crossed random-effects refit and two cluster-robust OLS specifications (clustered by agent and by item; Table S4) are reported as robustness anchors. The item fixed-effects estimate (cluster-robust SE by item) is the primary difficulty-controlled estimand; mixed-effects and cluster variants are sensitivity checks (SI *S7*). Participant identifiers are unavailable in the public release, so clustering is at the item level.

Trace-content features.

For each LRM trial on H-ARC and Cortes (INTUIT traces are "Restricted" in the public release) we compute four length-normalised features from the released `reasoning_trace`: self-doubt-marker density per 1,000 characters (`wait`, `actually`, `hmm`, `let me reconsider`, ...), self-correction-marker density (`scratch that`, `ignore that`, ...), word-level 5-gram repetition rate, and type–token ratio. For each feature we fit feature $\sim$ correct $+ \log w + C(\text{agent}) + C(\text{item})$ where w is trace length in words, cluster-robust SE by item. Full marker lists and the parser are in the OSF deposit (see *Data and Code Availability*).

Reproducibility and multiple comparisons.

Analyses are deterministic with seed `0xA12C`; full code at the OSF repository (see *Data and Code Availability*). Implicit family of tests and family-wise correction summary: SI *S8*.

3 Results

On the same items, humans and thinking LRMs agree on which ones are hard. They disagree on what to do about it. The registration test reproduces the de Varda et al. (2025) result on H-ARC: thinking-LRM trace length correlates positively with human reaction time (Spearman $\rho = 0.16$–0.30, all $p < .001$ across the five matched LRMs), while non-thinking DeepSeek-V3 does not ($\rho = -0.05$). The same five LRMs separate cleanly from the matched human samples on the allocation axis: every thinking LRM lands at $d > 0$ while the human point sits at $d = -0.10$. Registration position is uninformative about allocation position (Figure 1b). Two agents passing the same registration test can fail the allocation test in opposite directions.

3.1 The within-agent allocation gap on H-ARC

The H-ARC result is a sign reversal: humans spend less time on failures once item difficulty is controlled, whereas LRMs spend more tokens on failures. H-ARC carries the inferential weight: it has the largest matched LRM sample and the cleanest item-fixed primary test. The allocation gap on H-ARC has three layers, each shown as one panel of Figure 2. (i) Per-agent allocation. Every analysed thinking LRM produces longer log deliberation on wrong than on right trials. The well-powered range is $d = 1.47$–3.13 (DeepSeek-R1, GLM-4.5-Air-FP8, gpt-oss-120b, Qwen-QwQ-32B; all $n_{\text{LRM}} \geq 298$), while the matched human sample is at $d = -0.10$ (Figure 2a; Table S1). Per-bar failure-budget amplification annotations show that the well-powered LRMs spend at least a proportional share of their total compute on the trials they fail (FBA = 1.03–1.19). (ii) The direction reverses across systems on the same paradigm. Estimated marginal mean log deliberation by outcome (Figure 2b) shows a small *negative* slope on the human side (-0.08 log units, wrong $-$ right) and a large *positive* slope on the LRM side ($+0.78$). Held on their own scales, the two systems disagree on the sign of the correctness slope. (iii) The dissociation survives item fixed effects and several mixed-effects and cluster-robust variants (Figure 2c). The primary item

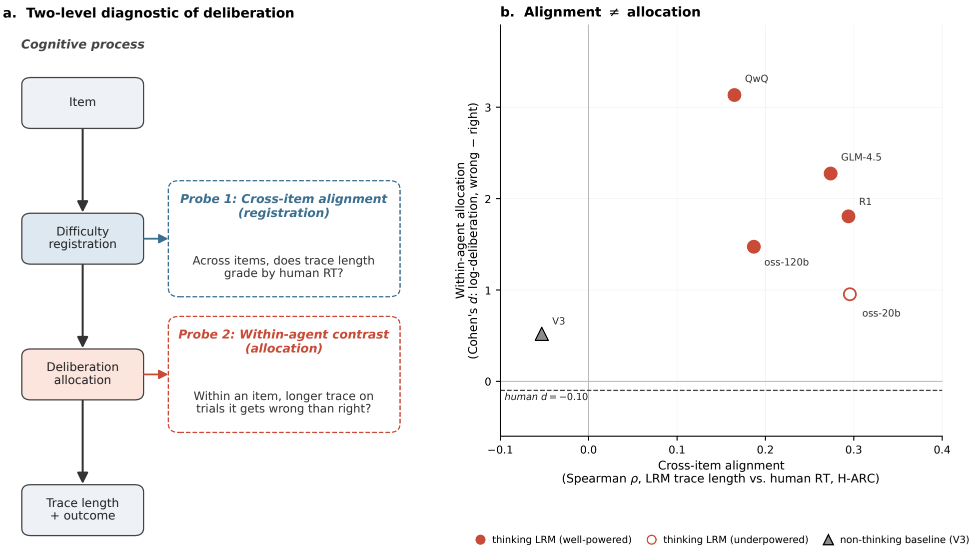

Fig. 1: **The two-level diagnostic.** (a) Schematic. An agent registers item difficulty (a perceptual or evaluative difficulty signal) and then allocates deliberation around that registered difficulty (a stopping or scheduling rule that decides whether to keep thinking on the item now in front of it). The cross-item alignment of de Varda et al. (2025) probes registration; the within-agent contrast introduced here probes allocation. (b) The two diagnostics on H-ARC. Horizontal axis: cross-item Spearman ρ between LRM reasoning-token length and human reaction time (registration). Vertical axis: within-agent Cohen's d on log deliberation, wrong − right (allocation). The thinking LRMs (five plotted; gpt-oss-20b shown as open-circle, parser-limited) all sit in the upper-right quadrant (positive on both diagnostics); the matched human reference is at $d = -0.10$ (dashed); the non-thinking V3 baseline is in the lower-left. Registration position is not predictive of allocation position. The vertical contrast is interpreted within each agent's native deliberation scale; seconds and reasoning tokens are not compared in absolute magnitude.

fixed-effects estimand absorbs item identity exactly and identifies the agent-type × correctness interaction purely from within-item variation: $\beta_{\text{LRM}} = -0.66$ log units ($p < .001$, 95% CI $[-0.81, -0.50]$, $n = 5{,}728$). Four sensitivity specifications triangulate the same negative interaction, with point estimates between -0.66 and -0.94 and all 95% CIs strictly below zero: item random intercept, crossed agent+item random effects, OLS with cluster-robust SE by item, and OLS with cluster-robust SE by agent (Figure 2c; Table S4). We treat the random-effects magnitude differences as expected shrinkage rather than substantive. A Mundlak (Mundlak, 1978) re-parameterisation that replaces item dummies with item-level mean correctness produces a still larger dissociation ($\beta_{\text{LRM}} = -0.79$, $p < .001$; within-item and between-item slopes differ at $p < .001$ in both agent groups; SI Table S2).

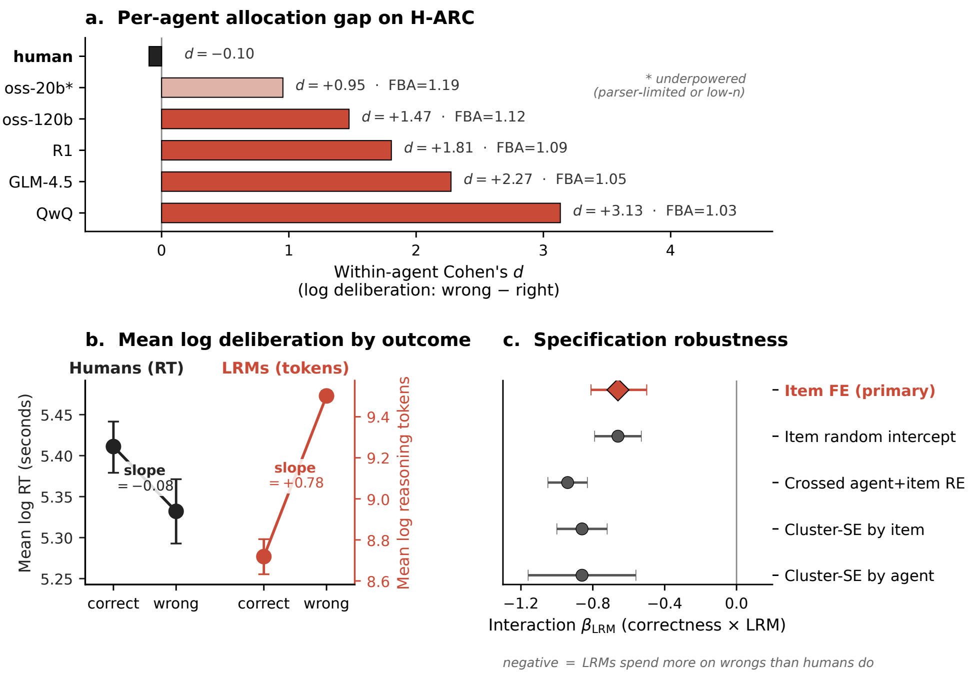

Fig. 2: **The within-agent allocation gap on H-ARC.** (a) Per-agent within-agent Cohen's d on log deliberation (wrong − right). The asterisk on gpt-oss-20b flags its parser-limited status; Qwen3-235B-Thinking is omitted here and reported in Table S1. *Failure-budget amplification* (FBA) is the share of LRM compute spent on wrong trials, divided by the share of trials that are wrong. (b) Estimated marginal mean log deliberation by outcome, with bootstrapped 95% CIs. Humans (left axis, log RT) descend on wrong trials; LRMs (right axis, log reasoning tokens) ascend. The two y-axes are scaled independently to align each system's within-agent change; cross-system magnitudes are not directly comparable. (c) Specification-robustness forest for the agent-type × correctness interaction; the diamond is the primary item-fixed-effects estimand (Table S4).

One feature of the LRM sample bears on the mechanism reading. The non-thinking DeepSeek-V3 baseline adds a useful constraint. V3 shows a positive d-ratio on Cortes ($d = 1.41$, 95% CI [1.06, 1.84]) and a directionally positive but underpowered d-ratio on H-ARC ($d = +0.52$, 95% bootstrap CI $[-0.22, +1.24]$, $n_{\text{right}} = 18$ at the 4.5% accuracy floor); V3 on INTUIT is too floored to estimate. V3 has no explicit reasoning trace and no thinking-targeted post-training. The V3 results therefore indicate that at least part of the wrong-trial length expansion in the thinking-LRM family is shared with non-thinking base-model generation under uncertainty, rather than being attributable exclusively to a reasoning-specific allocation policy. The data do not let us partition the share. The cogsci-side claim, however, is robust: the matched human samples on the same items show no comparable wrong-trial expansion.

3.2 Generalisation and the Cortes boundary

We treat INTUIT as the cross-paradigm replication test and Cortes as the boundary test. The former asks whether the H-ARC dissociation generalises to a structurally distinct multi-choice reasoning paradigm; the latter asks how far the allocation diagnostic survives when the task has a binary relational answer space and few LRM errors. INTUIT preserves the sign dissociation. Cortes preserves the agent-type contrast but reveals a boundary on the LRM-only within-item slope.

The same allocation contrast on the other two non-saturated paradigms (Figure 3) extends the H-ARC pattern in two informative ways. INTUIT (intuitive physics, multi-choice) is the clean cross-paradigm replication. Every analysed thinking LRM lands at $d > 0$ ($d = 0.59$–1.42 across six models) while the matched human sample is at $d = +0.07$, and the item-FE agent-type interaction is $\beta_{\mathrm{LRM}} = -0.41$ ($p < .001$). Cortes (binary relational reasoning) is a paradigm-dependent boundary. On Cortes, the marginal d-ratio and the item-fixed-effect slope diverge: the d-ratio remains wrong-longer overall (every thinking LRM at $d = 0.82$–1.96), but once item identity is absorbed, the pooled LRM correctness slope becomes positive (+0.31, opposite in sign to its negative slopes on H-ARC and INTUIT). We therefore treat Cortes as a boundary case rather than as a clean replication of the LRM-side mechanism. The agent-type interaction is nevertheless preserved ($\beta_{\mathrm{LRM}} = -0.97$, $p < .001$), because the human side is more strongly negative on Cortes ($d = -0.31$). On Cortes both systems sit on the engagement side of zero on the within-item slope, and the dissociation survives only because the human positive within-item slope is steeper than the LRM one. We do not interpret the Cortes inversion mechanistically here: per-LRM wrong-trial counts are small (5–8 per model) and the answer space is structurally distinct (binary versus generative or multi-choice). Per-paradigm item-FE coefficients and the underlying per-LRM interactions are reported in SI Tables S5–S6; per-LRM Cortes bootstrap intervals (Efron & Tibshirani, 1993) are wide but exclude zero for all six thinking LRMs (SI Figure S3).

At the per-LRM level the dissociation is rank-stable across paradigms: the same model that dissociates most strongly on H-ARC tends to dissociate most strongly on the other paradigms (Spearman $\rho = 0.77$–0.89 across the three paradigm pairs, $n = 6$ models per pair; Qwen-QwQ-32B is the strongest-dissociating model on every paradigm). With $n = 6$ this is descriptive, not inferential, but the direction is consistent with the dissociation reflecting a model-level (training-pipeline) signature rather than an item- or paradigm-specific artefact.

3.3 Inside the dissociation: human engagement, LRM length-on-uncertainty

The behavioural signatures suggest, but do not prove, different control policies: human time is partly gated by engagement, whereas LRM length tracks uncertainty-like trace features. A behavioural dissociation alone underdetermines mechanism. We use two convergent within-system probes (Figure 4) to sketch what is producing each side of the allocation gap, on its own terms. We use the labels "engagement-vs-abandonment" and "length-on-uncertainty" as mechanistic interpretations of the behavioural signatures

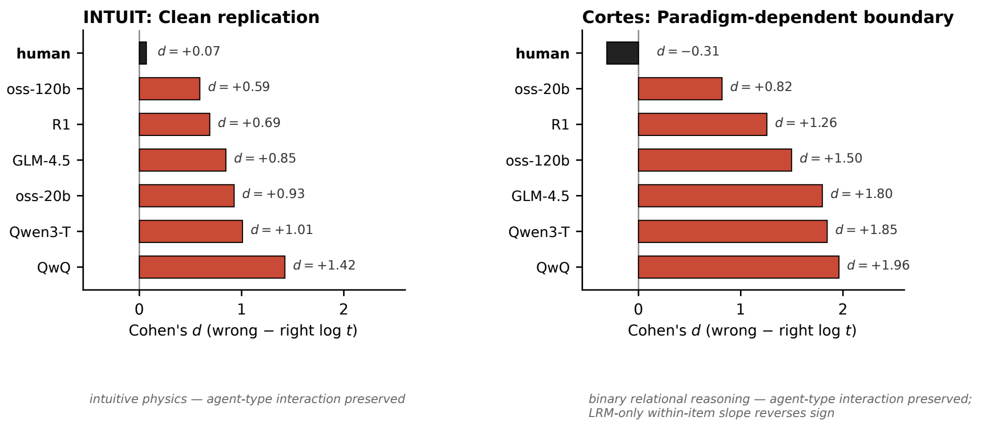

Fig. 3: **Cross-paradigm allocation gap.** Per-agent within-agent Cohen's d on log deliberation (wrong − right) for the two non-saturated paradigms beyond H-ARC. INTUIT (intuitive physics) is the clean replication; Cortes (binary relational reasoning) is a paradigm-dependent boundary in which the agent-type interaction is preserved but the LRM-only within-item slope reverses sign (see *Generalisation and the Cortes boundary*). H-ARC is the primary evidence and is shown in Figure 2a.

observed, not as direct evidence of internal architecture; the probes constrain the space of accounts but do not select a unique mechanism.

The two human numbers — a near-zero H-ARC d-ratio ($d = -0.10$, slightly negative; Figure 2a,B) and a *positive* within-item slope (+0.24 log units) — look at first like a contradiction but reflect different operations. The d-ratio mixes within- and between-item variance, so it inherits the cross-item difficulty confound that the within-agent contrast was designed to remove; once item identity is held fixed, the human slope is positive and reveals the engagement signature analysed below.

The positive within-item human slope on H-ARC (+0.24 log units) is not noise: it is the visible imprint of an engagement-vs-abandonment process separable from problem-solving itself. The H-ARC release preserves a per-trial action count (`Num_actions_attempt_1`, the number of grid edits the participant made), which decomposes the slope into engagement and disengagement components. Within the human wrong cell, the fast-wrong quartile shows median 16 grid actions while the slow-wrong quartile shows 79 (5× more); fast-wrong trials look like abandonment, slow-wrong trials like engaged-but-failed effort. The slope is concentrated on the harder items (bottom-quartile $\beta_{\text{correct}}{}^{\text{human}} = +0.32$, $p < .001$; top-quartile $+0.13$, $p = .05$), as one would predict if hard items disproportionately attract abandonment. Per-item RT coefficient of variation predicts per-item slope across 341 items with both right and wrong trials (Spearman $\rho = +0.27$, $p = 5 \times 10^{-7}$), and adding $\log(1 + \text{actions})$ as a covariate cuts the within-item slope nearly in half, from $+0.24$ to $+0.12$ (Figure 4a). The Mundlak decomposition corroborates: the human *between-item* correctness slope is -0.54 (items more humans solve have shorter mean RT), opposite to the within-item slope, exactly as predicted under abandonment between items and engagement within them.

The LRM side leaves a complementary footprint in the trace itself. Holding trace length, item, and agent fixed, wrong-trial chains on H-ARC contain 0.43 more self-doubt / hedging markers per 1,000 characters than right-trial chains ($\beta_{\mathrm{correct}} = -0.435$, $p = 7 \times 10^{-4}$; Figure 4b). The direction is the same in 5/5 thinking models. On Cortes (binary action space, fewer opportunities for verbal hedging) the asymmetry surfaces in lexical structure rather than in lexical hedging: wrong-trial Cortes chains show 7.4% higher 5-gram repetition ($\beta = -0.074$, $p = 7 \times 10^{-8}$) and 4.1 percentage-points lower type–token ratio ($\beta = +0.041$, $p = 10^{-3}$), with the predicted direction in 6/6 Cortes models. The two paradigms thus carry convergent rather than identical signatures: hedging surfaces in language-rich generative output, lexical contraction under a constrained binary action space. Both are consistent with a shared length-on-uncertainty mechanism. Surface markers are descriptive proxies, not mechanism; controlled truncation is the discriminating intervention (see Discussion; full regression in SI Table S7).

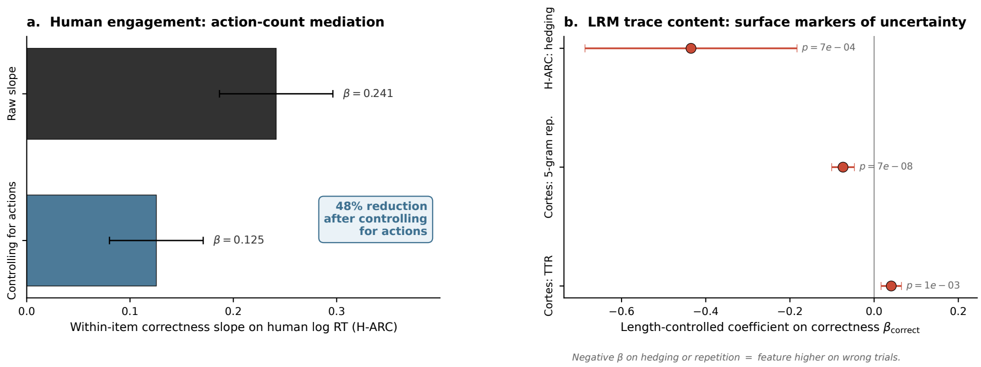

Fig. 4: **Two convergent behavioural probes of the proposed mechanism.** The labels are mechanistic interpretations of behavioural signatures, not direct evidence of internal architecture. (a) Human engagement decomposition. The raw within-item correctness slope on H-ARC human log RT is $\beta = +0.241$; controlling for $\log(1 + \text{actions})$ (the per-trial grid-action count) reduces it to $\beta = +0.125$, a 48% reduction. (b) LRM trace-content asymmetries: length-controlled coefficients on correctness for three features. Negative β on hedging or repetition means the feature is *higher* on wrong trials.

3.4 Robustness

The H-ARC allocation gap survives three further robustness checks (SI *S12*, Figure S2): a positive LRM d-ratio in nearly every difficulty quartile, an item-level human d-ratio of +0.60 that recreates the cross-item difficulty confound the within-agent contrast avoids, and an aggregated-cell re-estimation that returns the same $\beta_{\mathrm{LRM}} = -0.66$ (SI *S4*). The Supplementary Information is organised to separate inferential robustness checks from descriptive low-power cells: per-LRM H-ARC estimates (*S1*), aggregation against within-item human pseudo-replication (*S4*), specification variants (*S9*), multiple-comparison treatment (*S8*), and cross-paradigm item-FE regressions (*S10*) are reported separately so readers can audit the primary estimand without being asked to over-interpret descriptive cells.

4 Discussion

4.1 What the diagnostic reveals

The matched data show that registration and allocation can come apart: humans and thinking LRMs grade items by difficulty in the same order, yet allocate computation around that registered difficulty by different rules. A mechanism-level claim of shared reasoning is therefore not licensed by cross-item alignment alone. What follows is interpretive: behavioural dissociations under-determine mechanism, and the allocation patterns are *consistent with* the principles we sketch below, not proof of them.

On the human side, the within-item positive slope and its decomposition are consistent with an **engagement-vs-abandonment schedule**: a metacognitive evaluation of whether the item is worth pursuing. The signature is a roughly fivefold gap in grid actions between fast-wrong (abandoned) and slow-wrong (engaged-but-failed) trials, with action count mediating about half the slope. The Mundlak decomposition corroborates it: the *between*-item slope flips sign (more humans solve an item $\Rightarrow$ shorter mean RT on that item). The pattern is consistent with a value-sensitive engagement process: trials that end in success tend to show more sustained task engagement, whereas fast errors look more like disengagement or abandonment. Because participant identifiers are unavailable in the released data, this should not be read as a definitive demonstration of a trial-level stopping rule on the human side.

On the LRM side, the within-agent allocation gap on H-ARC, its replication across paradigms, and the trace-content asymmetries reported in *Inside the dissociation* are consistent with a **length-on-uncertainty policy**: when the model is uncertain it generates more reasoning tokens, with an uncertain relation to whether those tokens carry task-relevant computational value. Surface markers do not establish that the additional tokens lack computational value; they motivate an intervention. Controlled truncation is needed to separate useful late search from length expansion that continues after the answer has effectively stabilised.

The non-thinking DeepSeek-V3 control (*The within-agent allocation gap on H-ARC*) leaves two readings open: a *base-model* reading, on which the wrong-trial length expansion is a general property of next-token generation that thinking-LRMs inherit, and a *stratified* reading, on which thinking-LRMs add a reasoning-targeted component picked out by the trace-content asymmetries. The truncation experiment of *How to test this further* is the first cut at separating them. Either way the cogsci-side claim is robust: the human samples on the same items show no comparable wrong-trial expansion, and the two systems diverge not in *whether* difficulty registers but in the rule mapping registered difficulty to deliberation time. The next subsection makes that rule-level divergence formal.

4.2 A resource-rational synthesis

We read the dissociation through the lens of resource-rational analysis (Callaway et al., 2022; Gershman et al., 2015; Lieder & Griffiths, 2020) and its metareasoning antecedent (Russell & Wefald, 1991). Let $\mathrm{VOC}(t \mid \text{item}) = \mathbb{E}[\,\Delta\text{accuracy} \mid t \text{ more units}\,] - \kappa\, t$ denote the value of computation for t additional log-token units on a given item, with $\kappa > 0$ a monotone cost. The resource-rational stopping rule is to stop when marginal VOC ≤ 0. This rule does not predict that harder items *always* take longer: as item difficulty grows, the agent should engage harder when the marginal accuracy gain decays slowly (positive expected return on more t) and *abandon* when it vanishes (no return to be had). The within-item, outcome-conditioned slope is exactly the observable that separates these two regimes on the same item.

Under this lens, the human pattern (longer t on solved trials, short t on fast-wrong trials, with the engagement-vs-abandonment gap on slow-wrong trials) is consistent with the resource-rational rule under *any* monotone κ: extra t is allocated where expected marginal accuracy gain remains positive and withdrawn where it does not, and the agent's own subsequent solve/no-solve outcome is an informative (if noisy) revealed proxy for that expectation. The thinking-LRM pattern inverts this revealed mapping: extra t is allocated *ex ante* to items on which the model's own continuation behaviour ultimately reveals near-zero return. Reconciling this with a resource-rational stopping rule under *any* monotone κ requires one of two things: either the agent's expected-gain forecast fails to condition on success-relevant features of the item, or the objective being optimised is not item accuracy. Item fixed effects absorb shared item difficulty but not agent-specific affordances or model-specific calibration failures. We therefore cannot formally rule out the first horn; the data shift interpretive weight toward the second without excluding the first. The empirical contrast is therefore not a direct estimate of VOC, but a sign constraint on the mapping from registered difficulty to observable deliberation. On Cortes both systems land in the engagement regime (positive within-item slopes), so the VOC reading there is weaker and the dissociation survives only because the human positive slope is steeper than the LRM one. This is a conceptual rationality claim, not an empirical estimate of κ. LRM behaviour may well be optimal under a different objective (RL training signals, verifier rewards, length-conditioned preferences); the right reading is not "humans rational, LRMs irrational" but that the two systems are not implementing the same resource-rational deliberation policy even if each is sensible under its own objective. This is the formal version of the claim, made in *Why this matters* below, that a single "deliberation" construct papers over a real distinction.

4.3 Why this matters

The first consequence is methodological. The two-level diagnostic should be the default in future LRM-as-cognitive-model work: registration alone is not a mechanism claim, and any cross-item alignment argument should be paired with the within-agent allocation contrast on the same items.

The second consequence is metacognitive, and it is the most substantive cogsci claim the dissociation supports. The metacognition literature distinguishes second-order *monitoring* (verbalised uncertainty, hedging, confidence) from second-order *control* that acts on the first-order computation itself, terminating it when its expected value is low (Fleming & Lau, 2014; Nelson & Leonesio, 1988; Yeung & Summerfield, 2012). The matched dissociation suggests these two components are *decoupled* in current thinking-LRMs in a way they are not in the matched human samples. On the LRM side, monitoring signatures are present and even rise on wrong trials (hedging on H-ARC, lexical contraction on Cortes), but they appear weakly coupled to stopping: the model continues at length on the trials it ultimately fails. On the human side, a fast-wrong abandonment cell coexists with a slow-wrong engaged-but-failed cell on the same items, the hallmark of monitoring that does gate effort allocation. A single "deliberates longer when uncertain" description is therefore true of both systems at the cross-item level but misleading at the within-item level. The same surface phenomenon is consistent with a coupled monitor-controller in one case and with weaker coupling between uncertainty markers and stopping in the other. Whether thinking-LRMs have a separable stopping circuit at all, or whether their "stopping" is the autoregressive decay of generation, becomes the architectural question the data sharpen but cannot settle alone. The truncation experiment is the natural follow-up.

4.4 How to test this further

The clearest discriminating prediction is intervention. A controlled truncation manipulation, with repeated LRM samples at fixed token budgets $f \in (0, 1)$ on the same items where we observe the allocation gap, recovering $p(\text{correct} \mid t, \text{item})$, produces qualitatively different curves under the two principles (Figure 5). A length-on-uncertainty pattern would converge on its answer well before the trace ends, predicting a high-accuracy plateau under moderate truncation; a useful-search interpretation predicts a late, steep rise as truncation cuts into load-bearing computation. The two predictions separate most strongly in the $f \approx 0.5$–0.8 range. Alternative accounts remain open (a value-of-computation account under a non-task-aligned objective could still fit; SI *S5*), but truncation discriminates them at the level of the data they predict. The present paper establishes the dissociation empirically and identifies truncation as the discriminating intervention; running it is left to follow-up.

4.5 Limitations

Six limitations constrain what we can conclude from the data. First, **causal value of the extra LRM tokens.** The data are observational. We do not show that the extra wrong-trial tokens have zero or low causal value on those particular items; we show that their systematic allocation is the opposite of what the matched human samples do on the same items. This limitation is exactly why the truncation experiment is framed as the discriminating follow-up (see *How to test this further*), rather than as evidence already supplied by the present analyses. Because the public release contains a single observation per (item $\times$ model) pair (see Methods scope note), we cannot estimate a per-model $p(\text{correct} \mid t, \text{item})$ function; the outcome-conditioned within-agent

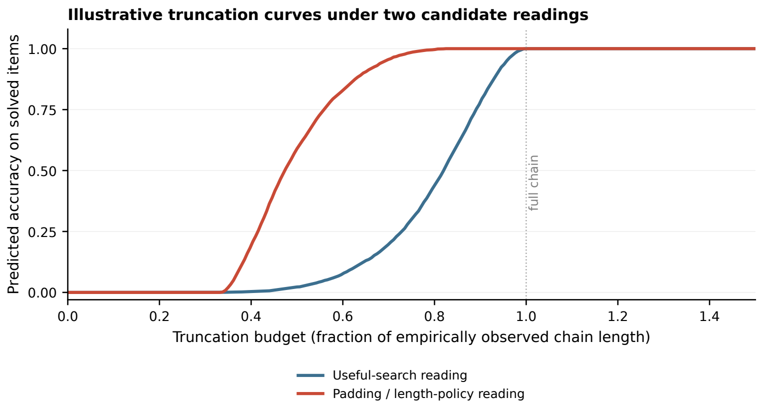

Fig. 5: **Illustrative truncation curves under two candidate principles** (illustrative simulation, not a fit). A useful-search interpretation predicts a late, steep accuracy rise as the truncation budget grows; a length-on-uncertainty / padding interpretation predicts an earlier rise and a broad high-accuracy plateau under moderate truncation. The curves diverge most strongly between $f = 0.5$ and $f = 0.8$.

contrast is the cleanest probe the data afford. The controlled-truncation experiment we describe is the natural follow-up; until it is run, claims about the computational value of the LRM tokens are interpretive. Second, **LRM scope.** The sample is six open-weight thinking LRMs from four organisations. Closed-frontier reasoning systems (e.g. o-series, Gemini-style frontier models) are not tested; the per-LRM rank-stability we report is descriptive at $n = 6$ and may not extend. Third, **human scope and identifiers.** "Humans" here means the human populations reported in the source release (de Varda et al., 2025; LeGris et al., 2025; Prunty et al., 2025), with their recruitment and selection criteria. The public release does not retain participant identifiers; participant-level differences in speed, ability, and engagement therefore cannot be separated from trial-level engagement, and the engagement-vs-abandonment reading is *consistent with* the action-count and Mundlak evidence rather than a definitive demonstration that humans implement a particular stopping rule. Fourth, **Cortes is a boundary case.** The LRM-only within-item slope on Cortes is positive (+0.31), opposite to H-ARC and INTUIT; the agent-type dissociation survives because humans show an even stronger positive slope, but the LRM unilateral within-item sign is paradigm-dependent. Fifth, **measurement.** LRM reasoning-token length and human RT are analogous, not interchangeable. Reasoning traces are not guaranteed reports of latent computation (Kambhampati et al., 2025; Samineni et al., 2025; Valmeekam et al., 2025); human RT decomposes into many sub-processes (Bogacz et al., 2006; Heitz, 2014; Ratcliff & McKoon, 2008). The within-agent contrast requires only outcome-conditioned structure in each system's own units; cross-system magnitude comparisons are not interpretable. Sixth, **architecture.** Behavioural dissociations constrain mechanism but cannot establish architecture. The length-on-uncertainty / engagement-vs-abandonment reading is a hypothesis the data are consistent with, not a claim the data prove.

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